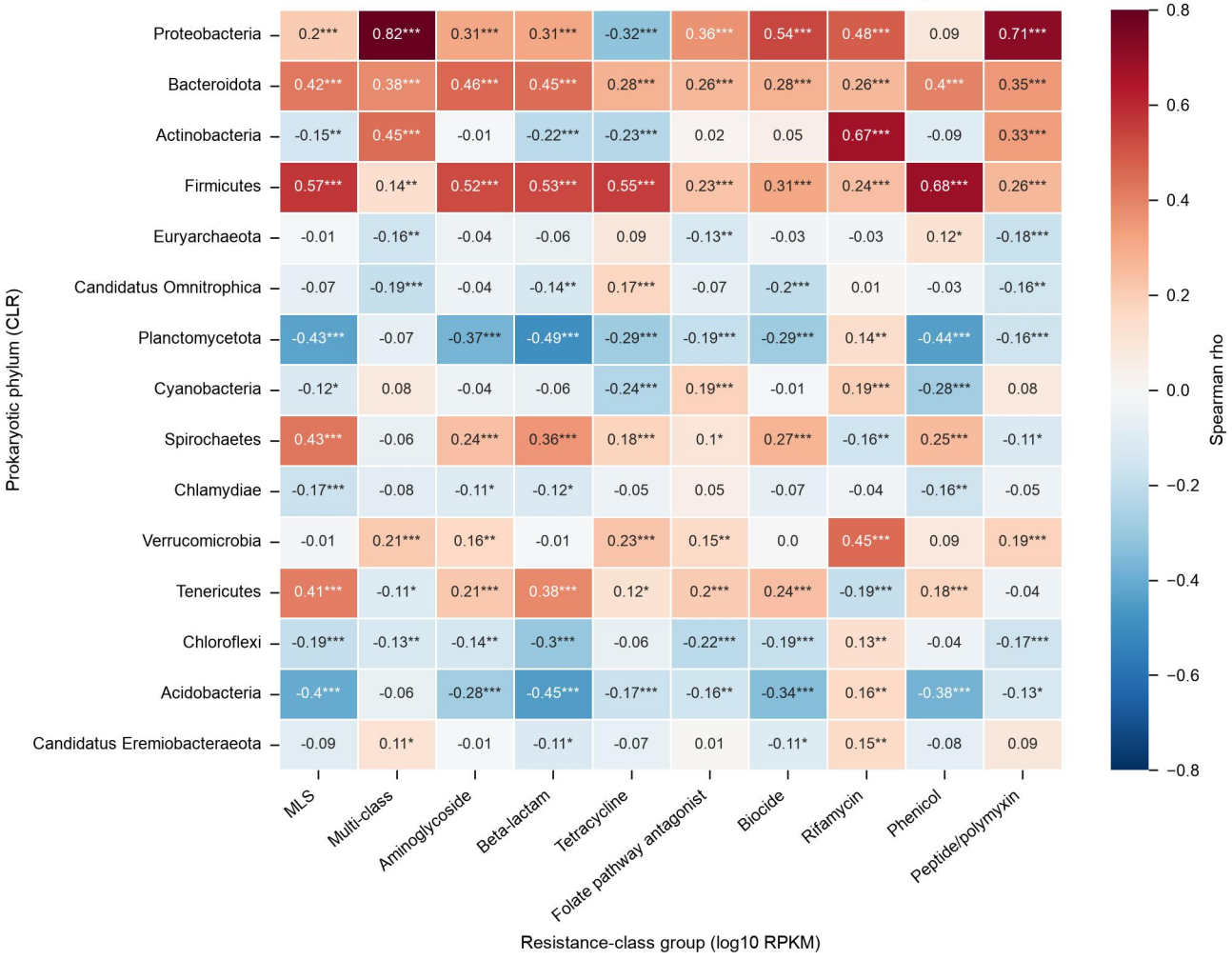

Phylum-resistome associations in DS02 wastewater (n = 447)
Prokaryotic phylum (CLR)
Proteobacteria 0.2*** 0.82*** 0.31*** 0.31*** -0.32*** 0.36*** 0.54*** 0.48*** 0.09 0.71***
Bacteroidota 0.42*** 0.38*** 0.46*** 0.45*** 0.28*** 0.26*** 0.28*** 0.26*** 0.4*** 0.35***
Actinobacteria -0.15** 0.45*** -0.01 -0.22*** -0.23*** 0.02 0.05 0.67*** -0.09 0.33***
Firmicutes 0.57*** 0.14** 0.52*** 0.53*** 0.55*** 0.23*** 0.31*** 0.24*** 0.68*** 0.26***
Euryarchaeota -0.01 -0.16** -0.04 -0.06 0.09 -0.13** -0.03 -0.03 0.12* -0.18***
Candidatus Omnitrophica -0.07 -0.19*** -0.04 -0.14** 0.17*** -0.07 -0.2*** 0.01 -0.03 -0.16**
Planctomycetota -0.43*** -0.07 -0.37*** -0.49*** -0.29*** -0.19*** -0.29*** 0.14** -0.44*** -0.16***
Cyanobacteria -0.12* 0.08 -0.04 -0.06 -0.24*** 0.19*** -0.01 0.19*** -0.28*** 0.08
Spirochaetes 0.43*** -0.06 0.24*** 0.36*** 0.18*** 0.1* 0.27*** -0.16** 0.25*** -0.11*
Chlamydiae -0.17*** -0.08 -0.11* -0.12* -0.05 0.05 -0.07 -0.04 -0.16** -0.05
Verrucomicrobia -0.01 0.21*** 0.16** -0.01 0.23*** 0.15** 0.0 0.45*** 0.09 0.19***
Tenericutes 0.41*** -0.11* 0.21*** 0.38*** 0.12* 0.2*** 0.24*** -0.19*** 0.18*** -0.04
Chloroflexi -0.19*** -0.13** -0.14** -0.3*** -0.06 -0.22*** -0.19*** 0.13** -0.04 -0.17***
Acidobacteria -0.4*** -0.06 -0.28*** -0.45*** -0.17*** -0.16** -0.34*** 0.16** -0.38*** -0.13*
Candidatus Eremiobacteraeota -0.09 0.11* -0.01 -0.11* -0.07 0.01 -0.11* 0.15** -0.08 0.09
MLS
Multi-class
Aminoglycoside
Beta-lactam
Tetracycline
Folate pathway antagonist
Biocide
Rifamycin
Phenicol
Peptide/polymyxin
Resistance-class group (log10 RPKM)
Spearman rho
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8